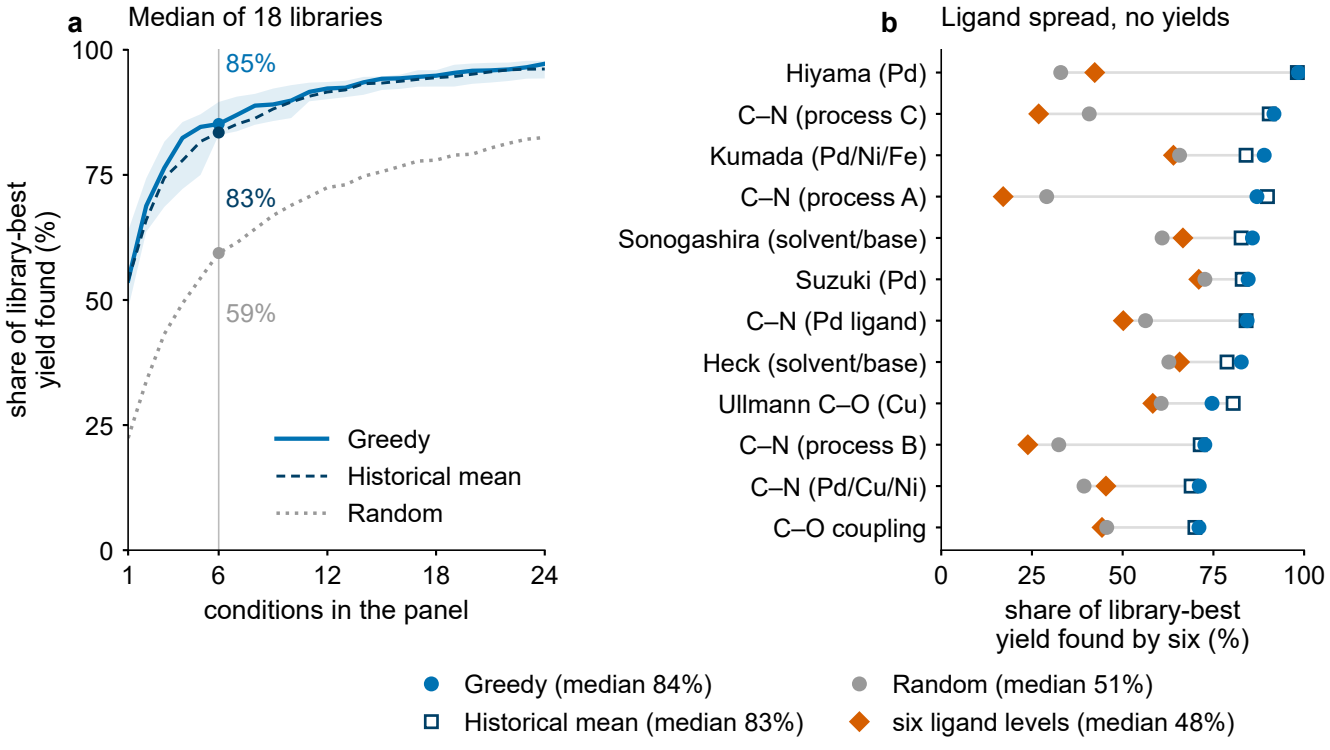
a
Median of 18 libraries
100
75
50
25
0
share of library-best yield found (%)
85%
83%
59%
Greedy
Historical mean
Random
1
6
12
18
24
conditions in the panel
b
Ligand spread, no yields
Hiyama (Pd)
C–N (process C)
Kumada (Pd/Ni/Fe)
C–N (process A)
Sonogashira (solvent/base)
Suzuki (Pd)
C–N (Pd ligand)
Heck (solvent/base)
Ullmann C–O (Cu)
C–N (process B)
C–N (Pd/Cu/Ni)
C–O coupling
0
25
50
75
100
share of library-best yield found by six (%)
Greedy (median 84%)
Historical mean (median 83%)
Random (median 51%)
six ligand levels (median 48%)